

Siba Smarak Panigrahi

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Education

Ph.D. in Computer Science <i>École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland</i>	2024 – Ongoing GPA: 5.5/6.0
M.Sc. (Thesis) in Computer Science <i>McGill University and Mila, Montréal, Canada</i> Thesis: Equivariance in the Era of Large Pretrained Models (Supervisor: Siamak Ravanbakhsh)	2022 – 2024 GPA: 4.0/4.0
B.Tech. in Computer Science and Engineering <i>Indian Institute of Technology, Kharagpur, India</i> Department Rank 2 in graduating batch of CSE students (Supervisor: Abir Das, Rameswar Panda).	2018 – 2022 GPA: 9.73/10
All India Senior School Certificate Examination (AISSCE) <i>Kendriya Vidyalaya Sangathan (KVS), India</i> Secured AIR 3, Rank 1 in Bhubaneswar Region. Among the top 0.1% of the 1.16 million candidates.	2018 98.6%

Research Interests

LLM Agents, AI Scientists/Co-scientists, AI for Scientific Discovery

Publications

[10] **Unsupervised Process Reward Models** ([Paper](#))
A. Gadetsky*, M. Kodryan*, **S. S. Panigrahi**, H. Guo, M. Brbić
Under review at Conference on Neural Information Processing Systems (NeurIPS) 2026.

[9] **HeurekaBench: A Benchmarking Framework for AI Co-scientist** ([Paper](#) | [Code](#) | [Website](#))
S. S. Panigrahi*, J. Videnović*, M. Brbić
International Conference on Learning Representations (ICLR) 2026.

[8] **BigDocs: An Open and Permissively-Licensed Dataset for Training Multimodal Models on Document and Code Tasks** ([Paper](#) | [Dataset](#) | [Website](#))
J. A. Rodriguez*, X. Jian*, **S. S. Panigrahi***, et al.
International Conference on Learning Representations (ICLR) 2025; NeurIPS 2024 - RBFM workshop.

[7] **SymmCD: Symmetry-Preserving Crystal Generation with Diffusion Models** ([Paper](#) | [Code](#))
D. Levy*, **S. S. Panigrahi***, S-O. Kaba, Q. Zhu, K. Lee, M. Galkin, S. Miret, S. Ravanbakhsh
International Conference on Learning Representations (ICLR) 2025; NeurIPS 2024 - AI4Mat ([spotlight](#)).

[6] **Improved Canonicalization for Equivariant Adaptation of Large Pretrained Models** ([Paper](#) | [Code](#))
S. S. Panigrahi, A. K. Mondal
CVPR 2024 - Workshop on Equivariant Vision ([EquiVision](#)).

[5] **Efficient Dynamics Modeling in Interactive Environments with Koopman Theory** ([Paper](#))
A. K. Mondal, **S. S. Panigrahi**, S. Rajeswar, K. Siddiqi, S. Ravanbakhsh
International Conference on Learning Representations (ICLR) 2024; EWRL 2023.

[4] **Equivariant Adaptation of Large Pretrained Models** ([Paper](#) | [Blogpost @ ICML'24](#) | [Code](#))
A. K. Mondal*, **S. S. Panigrahi***, S-O. Kaba, S. Rajeswar, S. Ravanbakhsh
Neural Information Processing Systems (NeurIPS) 2023, EquiVision (CVPR) 2024 ([spotlight](#)).

[3] **Identifying bot activity using topology-aware techniques** ([Patent](#))
G Choudhary, SI Rahaman, **S. S. Panigrahi**, P Bhutani, M Kilaru, K Singh, A Singhanian
US Patent 2023.

[2] **[Re]: Value Alignment Verification** ([Paper](#) | [Code](#))
S. S. Panigrahi*, S. Patnaik*
ML Reproducibility Challenge (MLRC) 2021; NeurIPS 2022 [Spotlight](#) and Journal Showcase Track.

[1] **Leveraging Pre-trained Language Models for Key Point Matching** ([Paper](#) | [Code](#))
M. N. Kapadnis*, S. Patnaik*, **S. S. Panigrahi***, V. Madhavan*, A. Nandy
EMNLP Workshop - Workshop on Argument Mining, 2021.

Industry Research Collaborations

Document Understanding with LLMs <i>Visiting Researcher Multimodal Foundation Models team</i> Developed permissively-licensed open dataset and improved document understanding capabilities of pretrained LLMs [8].	ServiceNow Research <i>Mar 2024 - Jul 2024</i>
Diffusion Models for Materials Generation <i>Supervisor(s): Siamak Ravanbakhsh, Santiago Miret, Michael Galkin</i> Trained conditional diffusion models for materials generation and crystal structure prediction task. Proposed meaningful representations for space groups and atom site symmetries for conditioning and interpretability [7].	Mila/Intel Labs <i>Sep 2023 - Aug 2024</i>
Contextual Bias in Visual Recognition Models <i>Supervisor(s): Abir Das, Dr. Rameswar Panda</i> Evaluated mAP and used GradCAM with state-of-the-art computer vision models to quantitatively and qualitatively determine the contextual bias in images containing exclusive and co-occurring biased pairs. Proposed shallow models to capture contextual bias and knowledge distillation approaches for automated bias mitigation to improve.	IIT Kharagpur/MIT-IBM Watson AI Lab <i>Apr 2021 - Dec 2021</i>
Improving Digital Marketing with Topological Data Analysis <i>Research Intern Supervisor(s): Iftikhar Ahamath Burhanuddin</i> Implemented Topological Regularization in LSTM Encoder-Decoder architecture to leverage topological information from customer navigation patterns and provide insights on customer sessions. Resulted in a published US patent [3].	Adobe Research <i>May 2021 - Jul 2021</i>

Academic Achievements & Honors

o Eastern European Machine Learning (EEML) Summer School (with travel scholarship)	(2026)
o AI for Science European Summer School (with travel scholarship)	(2026)
o Swiss AI Initiative PhD Fellowship within Swiss AI initiative projects	(2025)
o Amii Upper Bound bursary to attend Upper Bound AI conference	(2025)
o EDIC Fellowship for starting Ph.D. at EPFL	(2024)
o Mitacs Accelerate Fellowship at ServiceNow Research	(2024)
o Graduate Research Enhancement and Travel (GREAT) Award , McGill University	(2024)
o Oxford ML Summer School (OxML) (Representation Learning & Generative AI track)	(2024)
o Google's CS Research Mentorship Program (CSRMP) Class of 2023b	(2023)
o CIFAR Deep Learning + Reinforcement Learning (DLRL) Summer School	(2023)
o The Cornell, Maryland, Max Planck Pre-doctoral Research School (CMMRS)	(2023)
o Jamsetji Nusserwanji Tata (JN Tata) Scholar (endowment for higher studies)	(2022)
o Eastern European Machine Learning (EEML) Summer School (selected for poster presentation)	(2022)
o Research Week with Google (Computer Vision track) (1 of 150 selected students)	(2022)
o Prof. J.C. Ghosh Memorial Endowment Prize (highest CGPA after semester VI)	(2021)
o Indo-US Science and Technology Forum (IUSSTF) - Viterbi Award (1 of 15 awardees)	(2021)
o DAAD-WISE scholarship, University of Freiburg (declined)	(2021)
o Technology Alumni Association (Delhi Chapter) Award (highest CGPA after semester II)	(2019)
o Jagadis Bose National Science Talent Search Examination (Rank 2 of 173 awardees)	(2018)
o KVS Junior Mathematical Olympiad (Rank 6 in India; Rank 1 in Bhubaneswar region)	(2016)
o Exchange Student, Sakura Exchange Program in Science (1 of 90 selected students)	(2016)

Relevant Coursework

Mathematics - Linear Algebra, Calculus, Probability, and Statistics
Computer Science - Training Large Language Models, Geometry and Generative Models (GGM), Probabilistic Graphical Models, Network Science, Causal Inference and ML, Deep Learning, Machine Learning, Reinforcement Learning, Natural Language Processing (NLP), Information Retrieval, Principles of Programming Languages, Computer Networks*, Operating Systems*, Algorithms - I* & II, Cryptography & Network Security, Theory of Computation, Compilers*, Software Engineering* (* includes lab component)

Skills

o Programming Languages - Python, C, Java, L ^A T _E X, Verilog, MIPS
o Libraries - PyTorch, PyTorch Lightning, PyTorch Geometric, Jax, NumPy, Pandas, Huggingface, Hydra
o Web Development - Django, HTML, CSS, Bootstrap, PostgreSQL

Professional Services

- **Organizing:**
 - [ML Reproducibility Challenge \(MLRC\) 2023](#)
 - [CampusPulse](#) (sponsored **CA\$ 6000** GCP credits)
 - [Molecular ML Conference \(MoML\) 2024, MoML 2023](#)
- **Posters/Talks:**
 - Human Cell Atlas (HCA) General Meeting 2026, Boston (Poster, 2026)
 - International Conference on Learning Representations (ICLR) 2026, Rio de Janeiro (Poster, 2026)
 - Computer and Communication Sciences (EDIC, EPFL) Open House 2026, Lausanne (Poster, 2026)
 - International Conference on Learning Representations (ICLR) 2025, Singapore (Posters, 2025)
 - EquiVision workshop, Computer Vision and Pattern Recognition (CVPR) 2024, Seattle ([Talk](#), 2024)
 - Google Research, Bengaluru ([Talk](#), 2024)
 - International Conference on Learning Representations (ICLR) 2024, Vienna (Poster, 2024)
 - Conference on Neural Information Processing Systems (NeurIPS), New Orleans (Poster, 2023)
 - European Workshop on Reinforcement Learning (EWRL) 2023, VU Brussel (Poster, 2023)
 - NeurIPS 2022 Journal Showcase Track (and Spotlight lightning talk) (Poster & [Talk](#), 2022)
 - 50th Anniversary of School of Computer Science, McGill University (Poster, 2022)
 - KDAG Winter Workshop, lecture on Support Vector Machines ([Talk](#), 2020)
- **EDIC, EPFL Admission Committee:** Responsible for identifying strong PhD applicants (for Fall 2026)
- **Teaching Assistant (TA):** [CS401](#) (Head TA, EPFL, Autumn 2025, 600+ students), [CS250](#) (EPFL, Spring 2025)
- **Volunteering:** [DNetCV 2022](#) (CVPR-Workshop), EMNLP 2021
- **Reviewer:**
 - ICML 2026 (Gold Reviewer, Free registration)
 - NeurIPS 2025, 2024
 - ICLR 2026, 2025
 - [AI4Mat](#) (ICLR 2025, NeurIPS 2024 Workshop)
 - [GRaM](#) (ICLR 2026, ICML 2024 Workshop)
 - [NeurReps](#) (NeurIPS-Workshop 2024, 2023)
 - MoML (2024, 2023)
 - MLRC 2022 (Outstanding Reviewer)
 - DNetCV (CVPR-Workshop 2022)
- **Other:**
 - Member, Mental Health Committee, Mila
 - Head, Kharagpur Data Analytics Group (KDAG), IIT Kharagpur (organized [reading-sessions](#))
 - Head, Institute Wellness Group, IIT Kharagpur
 - Head Boy, Student Council, KV No.2 CRPF Campus, Bhubaneswar